

FREUID Challenge 2026 Short Report Draft

Submission ref **54511333**; public score **0.37009**. Lower FREUID score is better. This draft records the frozen method, validation evidence, release artifacts, and remaining package checks as of 2026-07-10.

1. Method Summary

The submitted system is a frozen two-branch image-forensics fusion for document-fraud detection:

$$\text{score} = 0.7 * \text{combined_v4_hgb}(\text{image}) + 0.3 * \text{convnext_tiny_logreg}(\text{image})$$

The submitted file contains 142,818 Kaggle sample-submission rows. The 7,821 IDs with locally available public-test images receive image-model scores; remaining sample-only IDs use the prior metadata fallback from earlier valid submissions.

2. Data And Validation

- Official archive extracted locally after Kaggle API throttling cleared.
- Matched train images: 69,352.
- Matched public-test images: 7,821.
- Working validation split: type+label-balanced 12,000 train / 4,000 validation, seed-controlled.
- Validation prioritized AUC, APCER at 1% BPCER, AuDET proxy, and calibration checks.

3. Branches

combined_v4_hgb: histogram-gradient-boosted classifier over photometric consistency, noise residual, JPEG/blocking, frequency, chroma, and reconstruction-lite features.

convnext_tiny_logreg: frozen ImageNet-pretrained ConvNeXt-Tiny encoder with a logistic classifier over embeddings. The backbone is not fine-tuned.

The original four-way fusion search included combined_v3_hgb and photometric branches, but the selected validation recipe assigned both zero weight. The frozen runtime therefore computes only the two nonzero branches.

4. Validation Results

Method	Accuracy	AUC	APCER @ 1% BPCER	AuDET proxy
combined_v3_hgb	0.8678	0.9407	0.2670	0.0595
photometric_logreg	0.8130	0.8880	0.3935	0.1121
combined_v4_hgb	0.8850	0.9584	0.2385	0.0417
convnext_tiny_logreg	0.8628	0.9457	0.2985	0.0550
frozen score fusion	0.8938	0.9661	0.2135	0.0341

The fusion improves both ranking and low-BPCER validation behavior over either branch alone.

Submission	Description	Public score
54481645	metadata binary canary	1.00000
54481898	metadata score baseline	0.98832
54481939	inverted metadata score	0.97578
54503265	first image fusion with metadata fallback	0.38042
54505114	larger combined_v4 image scores with metadata fallback	0.39303
54511333	frozen combined_v4 + ConvNeXt score fusion	0.37009

5. Reproducibility

- Repository: <https://github.com/DrStrangelOve/ai-image-forensics-comparison>
- Frozen artifact release: <https://github.com/DrStrangelOve/ai-image-forensics-comparison/releases/tag/freuid-freeze-2026-07-10>

- Runtime files: scripts/infer_freuid_frozen_stack.py, scripts/freeze_freuid_submission_artifacts.py, docker/freuid/Dockerfile, artifacts/freuid_2026/README.md, artifacts/freuid_2026/fusion_summary.json.

Input: flat image directory mounted at /data
Output: /submissions/submission.csv
Format: id, label
ID: filename stem
Label: fraud score in [0, 1]

- Local smoke test: 5 rows, mean score 0.132092.
- Max absolute difference versus previously materialized public fusion scores on the same IDs: 0.000337.
- Dockerfile scaffold is present; image build is pending because Docker Desktop Linux engine was unavailable locally.

6. Limitations

The sample submission contains many more rows than locally exposed public-test image files. The submission therefore uses image scores where images are present and metadata fallback elsewhere. This is valid for upload but means the public score partly reflects fallback behavior.

The validation split is source/type stratified but remains a local proxy. The weaker public score of the single combined_v4 candidate shows domain shift, so the frozen fusion was selected because it improves both local low-BPCER metrics and public score over the prior best.

7. Final Package Status

- Completed: valid Kaggle submission 54511333, public repo, frozen runtime release, score-aware lint/submit helpers, Docker inference scaffold.
- Remaining: build and run Docker image, export final PDF after any edits, post exactly one pinned discussion reply.